EVERY OPEN COVERING OF A PROFINITE SPACE IS REFINED BY A FINITE CLOPEN PARTITION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of a profinite space as a limit of finite discrete spaces over a directed set, with the projections forming a fundamental system of neighbourhoods. Invoked at the start of the counted proof (source lines 4430-4435). Supplies the finite quotients whose fibres give clopen sets.

Source reference. Original lemma lemma-profinite: sources/stacks-topology/pinned-original-source.tex, source lines 4351–4351. Navigation pointer only: the counted text refers to the Stacks item lemma-profinite, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a profinite space. Every open covering of X has a refinement by a finite covering $X = \coprod U_i$ with U_i open and closed.*

Proof. Write $X = \lim X_i$ as a limit of an inverse system of finite discrete spaces over a directed set I (Lemma lemma-profinite). Denote $f_i : X \to X_i$ the projection. For every point $x = (x_i) \in X$ a fundamental system of open neighbourhoods is the collection $f_i^{-1}(\{x_i\})$. Thus, as X is quasi-compact, we may assume we have an open covering

$$X = f_{i_1}^{-1}(\{x_{i_1}\}) \cup \ldots \cup f_{i_n}^{-1}(\{x_{i_n}\})$$

Choose $i \in I$ with $i \geq i_j$ for $j = 1, \ldots, n$ (this is possible as I is a directed set). Then we see that the covering

$$X = \coprod\nolimits_{t \in X_i} f_i^{-1}(\{t\})$$

refines the given covering and is of the desired form. □

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